Mapping Land-Cover Dynamics in Arid Regions Using Spectral Mixture Analysis and Representative Training Samples

Yanglin Cui, Chunjiang Zhao, Xuzhou Qu, Kai Ma, Yuchun Pan, Gaoxiang Yang, *Member, IEEE*, Qian Sun, and Xiaohe Gu

***Abstract*— Timely and accurate land-cover surveys are essential for understanding the drivers of global climate change and maintaining the carbon cycle. However, monitoring land-cover dynamics in arid regions presents challenges due to landscapes complexity and limited training samples. To address these challenges, this study proposes a novel approach that combines spectral mixture analysis (SMA) and adaptive strategies to automatically generate high-quality training samples for mapping large-scale land-cover dynamics in Guanzhong Plain (GZ), China. SMA was employed to develop new indices for detecting spatiotemporal changes and boundaries in land cover: the endmember abundance vegetation index (EAVI) and the heterogeneity detection index (HDI_{RMSE}). Subsequently, the spatial consistency region of multisource products was automatically generated using spatial superposition analysis, the morphological corrosion method, and HDI_{RMSE}. Once the initial training sample pool was created, it was optimized and updated through an adaptive sampling strategy to ensure a representative dataset. Additionally, by integrating the EAVI with CCD algorithm (CCD_EAVI), the spatiotemporally stable regions of land cover change were identified, enabling the automatic updating of annual training samples. Finally, a spatiotemporal consistency optimization approach was employed to improve the temporal stability of the time-series product. Based on independent validation samples, the overall accuracy of the land-cover mapping of the GZ in 2020 reached 97.92%, with an average mapping accuracy exceeding 96% over the past 32 years. Compared to benchmark products, these mapping results significantly improved spatial clarity, noise reduction, and classification accuracy. These findings highlight the proposed method's effectiveness for automated sampling and time-series land-cover mapping.**

***Index Terms*— Continuous change detection (CCD), google earth engine, land-cover, spectral mixture analysis (SMA), training samples.**

I. Introduction

LAND cover data are essential for understanding the complex interplay between human activities and environmental change, particularly in monitoring natural processes, securing food supply, and conserving the global carbon cycle [1], [2]. Since the Industrial Revolution, global climate shifts and human activities have significantly altered land-cover patterns. In arid regions, unauthorized land occupation, unsustainable development practices, and inefficient use of arable land have become increasingly prominent challenges [3], [4]. With the global population projected to exceed 9.7 billion by 2050 and per capita energy consumption rising, land-cover changes will accelerate [5]. In this context, timely and precise updates of land-cover information are crucial for effective disaster response, agricultural production enhancement, and carbon cycle maintenance.

With recent rapid advancements, remote sensing satellites, which possess extensive Earth observation capabilities, have become indispensable tools for real-time land-cover monitoring and predictive analysis of land cover [6], [7]. With the implementation of the open access policy of Landsat data archives in 2008, the emergence of cloud computing platforms, such as Google Earth Engine (GEE) and advancements in time-series analysis techniques, has enabled long-term monitoring of land cover changes using large remote sensing datasets [8]. The existing land cover product dataset spans the globe, with the spatial resolution having increased from 1 km to 1 m, and the monitoring cycle evolved from single temporal mapping to long-term dynamic monitoring [1], [9], [10]. However, significant discrepancies in accuracy persist in these products due to sensor issues regarding performance, classification scheme differences, and sample quality and quantity [1]. These issues are particularly evident in arid regions with high heterogeneity, resulting in inconsistent classification results. In addition, the lengthy update cycles of most

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products, approximately five years, limit their applicability in carbon emission assessments and spatiotemporal pattern analyses [11], [12], [13]. Thus, an urgent need exists for a more effective method to measure land-cover changes, enhance regional mapping accuracy, and enable real-time dynamic updates [12], [13].

Integration cloud computing with machine learning algorithms has significantly enhanced land-cover classification automation [14], [15], [16]. However, the effectiveness of supervised classification strategies depends heavily on the quality and quantity of training samples [17]. Traditional methods for obtaining training samples through manual visual interpretation are time-consuming, labor-intensive, and impractical for large-scale land cover mapping. Although some labeled datasets are available, they often rely on local sampling methods, leading to quantity, quality, and spatial resolution limitations [18] and restricting their application in areas significantly affected by natural changes and human activities [19]. In contrast, automatically extracting invariant features from the existing land-cover data to generate training samples greatly improves the automation ability of mapping. However, the inherent classification errors in each product may introduce invalid samples [16], [17], [20]. Additionally, using artificial thresholds based on phenological characteristics increases the likelihood of errors when migrating samples, particularly in arid, heterogeneous landscapes with complex planting systems [21]. Therefore, robust automatic sample generation and migration strategies are required to achieve efficient large-scale land cover mapping.

Dynamic monitoring of large-scale land cover changes commonly utilizes full-time-series and harmonic information from remote sensing images to construct detection models that identify abrupt and continuous changes trends on the Earth's surface. Over the past decades, key algorithms, such as Breaks for Additive Season and Trend (BFAST) [22], Landsat-based Detection of Trends in Disturbance and Recovery (LandTrendr) [23], and Continuous Change Detection and Classification (CCDC) [24], [25], have been instrumental in detecting land-cover changes [26]. However, most have focused on pixel-level change detection while often neglecting the contributions of different land cover elements and the sensitivity of spectral bands and vegetation indices to changes in saturated mixed pixels. To address the mixed-pixel issue in medium- and low-resolution satellite images, spectral mixture analysis (SMA) has been extensively applied [27]. For instance, Bullock et al. [28] utilized the Normalized Difference Fractional Index (NDFI) derived from the SMA method and the Continuous Degradation Detection (CODED) algorithm for extensive forest degradation assessment in the Amazon. Similarly, Chen et al. [29] applied the CCDC-SMA technique to monitor forest degradation in Georgia and combined CCDC with time-series NDFI analysis. Meanwhile, Cai et al. [30] introduced the Pure Forest Index (PFI) to analyze forest dynamics in fragmented landscapes using time-series satellite images. Despite these advances, researches have mainly focused on forest degradation within specific locales. The effectiveness of these methods for broader land-cover change in complex ecosystems remains uncertain.

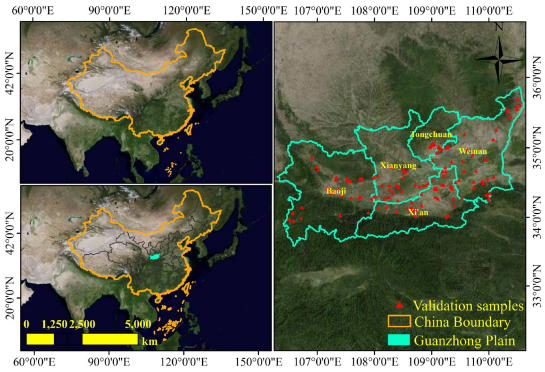


Fig. 1. Location of the study area, encompassing five cities in Shaanxi Province, China. The red triangles are 150 random points for land cover change time detection.

To enhance the accuracy and efficiency of large-scale land-cover change detection, redundant remote sensing features are typically eliminated based on the degree of feature contribution. However, this process is complex, cumbersome, and lacks generalizability [31]. Therefore, developing a limited set of representative features is crucial for land-cover change detection.

The current study proposes a novel framework for mapping large-scale land-cover dynamics maps in an arid region i.e., Guanzhong Plain (GZ), China, using SMA and adaptive sample generation strategies. The primary study objectives include: 1) developing sensitive features for subpixel land-cover change detection; 2) automating the generation and migration of training samples using SMA and CCD; 3) quantifying land-cover change and analyzing the spatiotemporal patterns of various land-cover types; and 4) assessing the performance of the produced maps with independent validation samples and multisource data products.

II. Study Area and Datasets

A. Study Area

The GZ, located in central Shaanxi Province, China, is an important region for wheat and corn production (Fig. 1). This predominantly flat area has elevations ranging from 323 to 3771 m (average: 1025.85 m). The region is characterized by a warm temperate, semi-humid continental monsoon climate with an average annual temperature of 12.9 °C, and a 215-day frost-free season. Moreover, most rainfall occurs between July and September [32]. Historically, the GZ has been a typical arid agricultural region with complex and rapidly changing farmland systems [33]. However, rapid urbanization has exacerbated the dynamic changes in the area's land cover.

B. Satellite Datasets

For this study, all available Landsat Collection 2 surface reflectance data from 1990 to 2022 covering the GZ were

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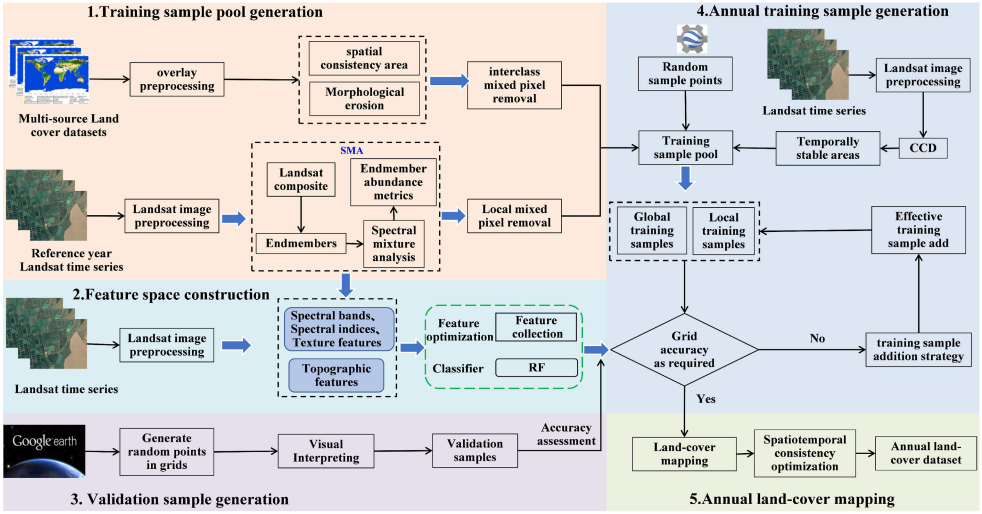


Fig. 2. Schematic of the methods for automatically generating training samples and time-series land-cover mapping by combining spectral mixing analysis, the adaptive sampling strategy, and change detection.

utilized on the GEE platform (Fig. S1). To reduce the computational burden and simplify the methodology, utilizing uncalibrated Landsat time series data was applied [29]. To mitigate issues related to cloud contamination and the limited temporal resolution of Landsat reflectance imagery, the dataset for the target year encompassed reflectance images from the preceding and succeeding years to ensure comprehensive domain coverage. Furthermore, all Landsat images underwent processing using QA bands through the Function of mask (Fmask) algorithm on GEE to eliminate clouds and cloud shadows. Beyond Landsat data, the Shuttle Radar Topography Mission (SRTM) dataset ("USGS/SRTM GL1_003"), offering a 30-m spatial resolution, was employed to derive three topographic variables: slope, aspect, and elevation.

C. Land-Cover Products

Three land-cover products with a resolution of 30 m—GLC_FCS30 [13], CLCD [19], and FROM-GLC30 [34]—were used to generate the sample training pool. To better reflect the specific land-cover characteristics of the GZ, the land-cover classification system was adapted to incorporate regional nuances (Table S1). Considering that the wetlands were relatively scarce and primarily located near river channels, they were merged into the water category. Additionally, the accuracy of cropland distribution was compared via CACD [21].

III. Methodology

Fig. 2 presents a comprehensive framework for automated sample generation and time-series land-cover mapping. This framework integrates multisource datasets with the SMA method, continuous change detection (CCD) algorithm, and hexagonal adaptive sampling strategy. The process comprises four distinct steps: 1) utilizing SMA in combination multisource datasets to generate the automated training sample pool; 2) creating a change detection index feature set through SMA and leveraging hexagonal adaptive strategies for the automatic enhancement and updating of effective samples; 3) applying the CCD algorithm with the entire suite of Landsat time-series imagery to identify spatiotemporal stable regions for automated sample updates; and 4) implementing spatiotemporal consistency verification optimization techniques to improve the stability and accuracy of time-series classification products.

A. Spectral Mixture Analysis

SMA enables the representation of all pixels within Landsat images through the physical abundance of spectral endmembers [27]. In the GZ, five endmember types were identified to unmix all cloud-free Landsat images: green vegetation (GV), nonphotosynthetic vegetation (NPV), soil, shadow, and cloud. High-resolution Google Earth imagery from corresponding periods assisted in selecting candidate pixels within Landsat images. The central pixel of a uniform land-cover type within a 3 × 3 window was selected as the endmember through visual interpretation. Across different landscapes, 50 potential endmembers were identified for GV in pristine GV zones, NPV in areas covered with dry leaves in October, soil in bare soil regions, and cloud in areas with clear cloud presence, respectively. The spectral reflectance of the final endmembers was averaged from that of candidate endmembers. The reflectance for the final shadow endmember was set to zero across all wavelengths (Fig. S2).

A fully constrained least squares linear SMA model [35] computed the fractional composition of each Landsat pixel's endmembers (1), with nonnegative abundance values totaling 1 (2)

$$F_n = \sum_{m=1}^{M} f_m a_{m,n} + \varepsilon_n \tag{1}$$

$$\sum_{m=1}^{M} f_m = 1, \quad (0 \le f_m \le 1) \tag{2}$$

where F_n denotes the reflectance of a pixel in the nth band, $a_{m,n}$ is the reflectance of endmember m in the nth band,

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f_m signifies the abundance fraction of the mth endmember, n is the spectral band count, m is the endmember count, and ε_n is the pixel's residual value in the nth band.

The SMA efficacy was assessed by calculating the root mean square error (RMSE) of the annual median synthetic images for subregions as follows:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{n=1}^{N} \varepsilon_n^2}{N}}. \tag{3}$$

The average RMSE values for 1987, 2005, 2010, 2015, 2020, and 2022 were 0.0039, 0.0044, 0.0038, 0.0064, 0.0049, and 0.0058, respectively. Fig. S3 shows that the spatiotemporal consistencies of the five endmembers align well with the spectral characteristics of the land-cover observed in the Landsat images.

B. Development of the EVAI

1) EAVI Principle: The spectral reflectance of land surface features is intrinsically linked to the composition and relative proportions of components within a pixel. In the process of unmixing Landsat images, the spectral value of each pixel is depicted as the cumulative product of the spectral reflectance of various endmembers and their respective abundances [27]. Typically, the spectral indices of a pixel are influenced by the proportion of the principal endmember within a mixed pixel, while the variations in lesser constituents may not markedly affect the spectral reflectance. Thus, formulating sensitivity spectral indices specific to distinct endmembers and detecting changes in the composition of dominant endmembers are crucial for enhancing the precision of land-use change detection. To secure a comprehensive set of Landsat images encompassing the study area and reduce phenological interferes on the generalizability of sensitivity indicators, all Landsat images from 2019 to 2021 were employed to select endmember sensitivity indicators. Initially, median composite images for each quarter of the reference year were generated to calculate commonly utilized vegetation indices (Table S2) and gather information on endmember abundances for pixels. Subsequently, 5000 pixels were randomly selected and analyze the Pearson correlation between the endmember abundances of these pixels and six spectral bands and 12 vegetation indices (Fig. 3). Sensitivity indicators for various endmembers were determined based on correlation strengths, identifying EVI, NDBI, SWIR1, MNDWI, and Red as the indicators for GV, NPV, shade, cloud, and soil, respectively. To mitigate the impact of mixed pixels and improve the accuracy of land-use change detection, a novel change detection index, i.e., the endmember abundance vegetation index (EAVI), was developed by the following equation:

$$\mathrm{EAVI}_m = \mathrm{VI}_i \times A_m \tag{4}$$

where VI_i represents the ith spectral band or vegetation index exhibiting the strongest correlation with the mth endmember abundance; A_m denotes the abundance of the mth endmember within pixel A. VI acts as a sensitivity indicator directly derived from Landsat images to endmembers, whereas A_m is determined using SMA for image unmixing.

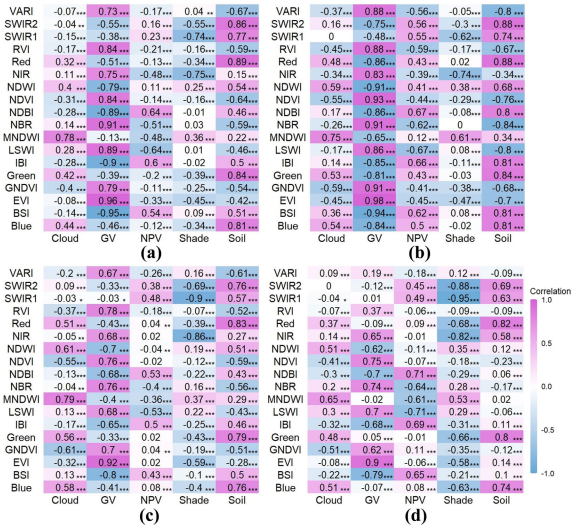


Fig. 3. Correlations between the abundance of GV, NPV, Cloud, Shade, and Soil endmembers with common spectral bands and vegetation indices during different periods. (a) March–May, (b) June–August, (c) September–November, and (d) December–February.

TABLE I
EIGHT DIFFERENT FEATURE TYPE COMBINATIONS COMBINED INTO THE CCD ALGORITHM

Feature Type	Input Metric	Reference
EAVIs+NDFI	$\mathrm{EVI_{GV}}$+$\mathrm{NDBI_{NPV}}$+$\mathrm{SWIR1_{Shade}}$+$\mathrm{Red_{Soil}}$+ $\mathrm{MNDWI_{Cloud}}$+NDFI	/
EAVIs	$\mathrm{EVI_{GV}}$+$\mathrm{NDBI_{NPV}}$+$\mathrm{SWIR1_{Shade}}$+$\mathrm{Red_{Soil}}$+ $\mathrm{MNDWI_{Cloud}}$	/
VIs	EVI+NDBI+SWIR1+Red+MNDWI	/
Endmembers	GV+NPV+ Shade+ Soil +Cloud	/
EAVIs_SMA	$\mathrm{EVI_{GV}}$+$\mathrm{NDBI_{NPV}}$+$\mathrm{SWIR1_{Shade}}$+$\mathrm{Red_{Soil}}$+NDFI	/
CCD_SMA	GV+NPV+ Shade+ Soil +NDFI	[29]
CCD	Green+Red+NIR+SWIR1+SWIR2	[24]
CODED	NDFI	[28]

2) Continuous Land-Cover Changes Mapping With the CCD_EAVI Algorithm: The CCD algorithm employs the harmonic regression model coefficients of all spectral bands within dense time series to predict future observation values [25]. By comparing the residuals between predicted and consecutive observed values, the algorithm identifies spectral changes at the pixel level. It also examines discontinuities in land-surface spectral reflectance throughout the time series. If the residual surpasses a predetermined statistical threshold, a fresh regression model is established and fitted; this process continues until the series concludes. To affirm the EAVI index's superiority in land-cover change detection, its performance was evaluated in conjunction with the CCD algorithm [29] and eight feature combinations (Table I). Unlike the original CCD algorithm, which used spectral bands and observations throughout the year, Landsat images from day 60 to day 270 were used to minimize the model's risk of being affected by snow.

C. Feature Space Construction

Optimizing the feature space is essential to develop a consistent and functional classification system. This process improves the efficiency of machine learning algorithms and

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reduces computational requirements while minimizing the risk of overfitting the model. During feature set construction, the EAVI—derived from vegetation indices and endmember abundance metrics—combined with the CCD algorithm accurately identified breakpoints in land-cover changes and was more adept at detecting land-use changes than traditional approaches using Landsat bands, spectral indices, and endmember indicators (Figs. S4 and S5). By analyzing spectral decomposition maps across various subregions, the NDFI emerged as particularly effective in distinguishing vegetation and impervious surfaces (Fig. S3). Accordingly, six endmember abundance features were selected, including five EAVIs (EVI_{GV}, $NDBI_{NPV}$, $SWIR1_{Shade}$, Red_{Soil}, and $MNDWI_{Cloud}$) and the NDFI. To capture the spectral mean and variability, metrics, such as average, standard deviation, amplitude, and 10th, 25th, 50th, 75th, and 90th percentiles for each spectral band were computed using a composite metric approach. Incorporating texture features can reportedly significantly improve the accuracy of classification maps [36]. Therefore, texture features of the EVI_{GV} abundance change detection indicator, which exhibits sensitivity to vegetation information, were integrated based on the gray level co-occurrence matrix (GLCM) methodology. Six texture variables of EVI_{GV} from annual median composite images were included: variance, homogeneity, contrast, dissimilarity, entropy, and correlation. In addition to the indicators derived from Landsat imagery, slope, aspect, and elevation data derived from the SRTM were incorporated into the feature set.

D. Automatic Generation of Training Samples

1) Training Sample Pool Generation: To ensure the temporal and spatial representativeness and stability of the training sample pool, spatial overlay analysis techniques were employed to assess the spatial consistency of three land-cover products (GLC_FCS30: 2010, 2015, and 2020, CLCD: 2010, 2015, and 2020, and FROM-GLC30: 2017) under a unified classification system. The maximum period covered by these products is ten years. Pixels with consistent classifications across the three products were deemed high-confidence pixels and selected as preliminary sample areas. Due to the dispersed and fragmented nature of grassland, water, and bare land in the study area, areas consistently classified by any two products were selected as preliminary sample areas for these land-cover types.

Due to the potential high error rate from varying data sources and classification schemes, especially in regions with high land-cover heterogeneity, and the vulnerability of individual pixels to noise, several measures were implemented to ensure the reliability and stability of selected samples. The optimal focal radius values for different land-cover types were determined using focal statistics [17]; erosion operations were performed on these types with respective radius values. Nonetheless, the challenge of "different objects with the same spectrum and the same object with different spectra" persisted due to systematic issues, such as data source resolution and classification schemes. Therefore, preliminary sample areas may include within-class classification errors even after erosion. Given the expected similarity in composition of the same type of land object elements within a small area and inspired by the RMSE metric, a new indicator, heterogeneity detection index (HDI_{RMSE}) was developed (5) to enhance the detection of internal area heterogeneity and minimize the chance of including samples in areas with spectral confusion (Fig. S6)

$$HDI_{p,RMSE} = \frac{\sum_{i=1}^{n} \sqrt{\frac{\sum_{j=1}^{m} D_j^2}{m}}}{n} \tag{5}$$

where HDI_{RMSE} is the aggregate of differences in endmember abundance between the target pixel and its neighbors in the target year's median composite Landsat image post-unmixing; and D_j is the difference in endmember abundance indicator j between the target pixel and its surrounding pixels. Since five endmember indicators were selected to unmix the Landsat images, the total number of endmember abundances m was set to 5, and the count of neighboring pixels within the window (n) was assigned a value of 8.

The HDI_{RMSE} value for each pixel within a 3×3 pixel moving window was calculated for the median composite image of the target year. Areas with HDI_{RMSE} values <0.08 in the target year were retained in the sample training pool for land-cover classification.

2) Hexagonal Adaptive Sampling Strategy: In geographic data analysis, hexagonal sampling outperforms quadrilateral sampling by providing more uniform and regular analysis units with equal area distribution and clearer adjacency relationships. This significantly improves sample collection efficiency and accuracy within the study area. To maintain an adequate and evenly distributed quantity of training samples, a hexagonal adaptive automatic sampling method was developed (Fig. 4).

1) *Initial Sample Determination* The optimal initial number of samples was determined by analyzing the area proportions of different land-cover types in the existing products (Fig. S7). Through stratified sampling, the relationship between sample size and accuracy was evaluated starting with 1000 points and incrementally increasing by 500, ultimately setting the initial sample number at 7500 (Fig. S8).
2) *Adaptive Sample Addition Strategy:* This method enhances training samples by exploiting discrepancies between global and local classification outcomes. Global classification utilizes all initial samples to create a feature set for modeling and classifying the entire study area. In contrast, local classification focuses on the feature similarity in adjacent areas due to the study area's vast scale, with data from each grid and its six neighbors used for model training. This process is repeated across all grids. Postclassification, areas inconsistent between global and local classifications were pinpointed via overlay analysis. These regions were intersected with the training sample pool to generate areas for sample addition, excluding areas where global classification results match the training sample pool. Additional samples were introduced through stratified sampling based on land-cover type proportions within these areas, with 500 samples added each iteration. The criteria for initial sample proportions and quantities

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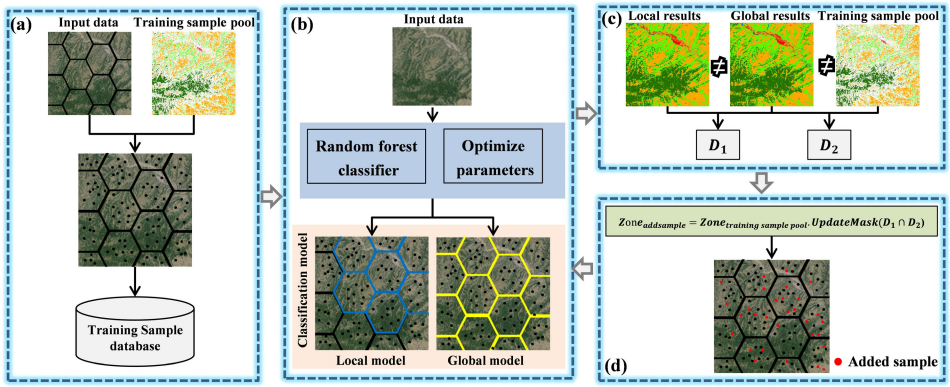


Fig. 4. Schematic of the adaptive sample addition strategy. (a) Automatic collection of training samples for the training sample pool. (b) Construction of global and local RF classifiers. (c) Overlay analysis of the classification results and training pool. (d) Construction of the effective sample addition strategy.

remain constant throughout subsequent additions. The process continues until the accuracy metric meets the requirements: samples were typically added in three cycles.

E. Automatically Update the Annual Training Sample Pool

Integrating the EAVI and CCD algorithm facilitates automated annual training sample updates by identifying stable areas over time. However, updating samples directly from stable areas within the 1990–2022 series is challenging due to the complex spectral characteristics of targets, particularly croplands affected by cropping systems and phenological factors. This can result in inadequate training sample numbers and distribution in unstable cropping zones, notably in arid regions. To counter this, the stable areas from the nearest year (2010 or 2020) to the reference year were used to automatically update the training sample, ensuring the representativeness and minimizing cropping system impacts on automated updates.

F. Classification and Spatiotemporal Consistency Optimization

The random forest (RF) method is favored for its simplicity, high accuracy, ability to handle high-dimensional data without necessitating feature selection, and compatibility with parallel computation [37]. RF operates as an ensemble learning model and utilizes multiple decision trees for training and prediction; its accuracy slightly increases as the number of trees grows. The optimal tree count, selected from 10 to 500 in increments of 10 based on the classification accuracy, and other classifier parameters adhere to the GEE defaults. Despite implementing a subpixel-based classification feature library using SMA, automatically updating and supplementing training samples with the hexagonal adaptive algorithm for long-term land-cover mapping, challenges persisted in addressing potential systematic classification errors from transitional land-cover types, akin to those encountered by Li et al. [12]. To mitigate these issues, a temporal consistency optimization method incorporating spatiotemporal neighborhood information was utilized (6), which enhanced the reliability of land-cover classification products through optimized homogenous intervals

$$\text{Prob}_{j,t} = \frac{1}{n}\left[\sum_{x-1}^{x+1}\sum_{y-1}^{y+1}\sum_{t}^{t+2} I(L_{j,t} = L_{i,t})\right] \tag{6}$$

where $\text{Prob}_{j,t}$ is the probability of pixel j at spatial position (X, Y) and time t being a homogeneous pixel. According to Li et al. [12], a higher $\text{Prob}_{j,t}$ value within a $3 \times 3 \times 3$ spatiotemporal filtering window suggests a higher probability of homogeneity, reducing systematic errors caused by classification. () is an indicator function equation for the state between two pixels, with $L_{j,t}$ and $L_{i,t}$ representing the center pixel of the filtering window and its adjacent pixels, respectively. This study adhered to the recommendation of using a Prob threshold of 0.5 to identify classification errors; if $\text{Prob}_{j,t}$ was <0.5, the pixel was further corrected based on the types of surrounding pixels [38].

G. Accuracy Assessment

The numbers of 100 random points within each hexagon covering the GZ on GEE were visually interpreted using Landsat NDVI and MNDWI time series curves; characteristics, such as color, shape, and texture, were assessed. Considering the fragmented distribution of bare land, grassland, and water bodies, corresponding samples were added through visual interpretation. Due to the impracticality of interpreting validation samples annually over a long time series, datasets for 1990, 2000, 2010, and 2020 were visually interpreted (Table S3). The confusion matrixes, including the user accuracy (UA), producer accuracy (PA), overall classification accuracy (OA), and F1-score [39], were applied during the accuracy evaluation.

To test the accuracy of CCD_EAVI (EAVIs + NDFI) for land-cover change detection in time, the year and type of the first change were visually interpreted at 150 random land-cover change points by combining historical Landsat images from 1990 to 2022 and high-resolution images from Google Earth (Fig. 1)

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IV. Results

A. Applicability Analysis of the EAVI

To evaluate the efficacy of the EAVI in detecting land-cover changes, the CCD algorithm was employed to generate maps highlighting changes within the region from 1990 to 2022 (Fig. S5). The combination of endmember indicators outperformed traditional spectral bands and indices, particularly in identifying changes in roads and fragmented landscapes [Fig. S5(C), (D), and (G)]. The incorporation of the EAVI indicator, derived from spectral bands, indices, and endmember abundance metrics, markedly enhanced the detection of complete land-cover change areas [Fig. S5(B)–(D) and (E) and (F)]. A single NDFI indicator could not effectively identify the complete land-cover change area [Fig. S5(H)]. Furthermore, the addition of the NDFI and $MNDWI_{Cloud}$ indicators substantially improved the delineation and detection capabilities for smaller patches [Fig. S5(A), (B), and (E)].

Table II demonstrates the accuracy of CCD_EAVI in detecting land-cover change years. Temporally, 94% of the land-cover change events matched the year of disturbance, and 97% of the land-cover change events occurred within one year before or after the year of disturbance. The average deviation in years was 0.07, and the mean absolute deviation was 0.13 years.

B. Improved Representativeness of Collected Training Samples

An adaptive sampling strategy can significantly enhance the representativeness of training samples collected from the training pool. In this study, the t-distributed stochastic neighbor embedding (t-SNE) approach [40] was used for nonlinear projection in high-dimensional space. The feature set from Section III-C resulted in a 2-D feature space distribution of samples from six land-cover types (Fig. 5). Initially, clear boundaries existed between the six land-cover samples clusters. As samples were added using the adaptive sampling strategy, the intersecting areas of clusters increased. Hence, easily confused samples were extracted using this method, further improving the representativeness of the collected samples.

C. Spatiotemporal Pattern of Land-Cover Change in the GZ

Utilizing the combined EAVI and NDFI indicators, change detection and frequency mapping were conducted for the GZ covering from 1990 to 2022 (Fig. S9). Spatial analysis revealed distinct land-cover dynamics across the region characterized by notable variations in change intensity, with predominantly high values in the north and east and lower values in the south and west. In particular, the Weinan area witnessed the most extensive land-cover alterations, primarily across its southern and northern mountainous agricultural zones, with significant changes also noted in the central-northern Xianyang and eastern Xi'an areas. The counties of Pucheng, Chengcheng, and Baishui in Weinan exhibited the highest frequency of land-cover changes, predominantly ranging between two and three instances, whereas the Yellow and Wei River channels experienced changes exceeding five instances. Temporal analysis provided an expansion pattern of land-cover changes that

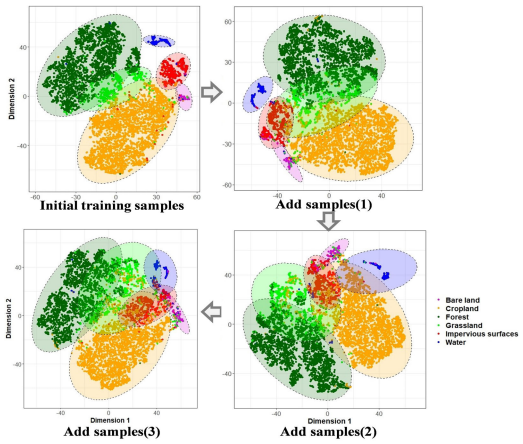


Fig. 5. Adaptive strategy for improving the representativeness of samples for classification. The nonlinear projection of a feature set from a high-dimensional space into a 2-D space using the t-SNE approach for visualization.

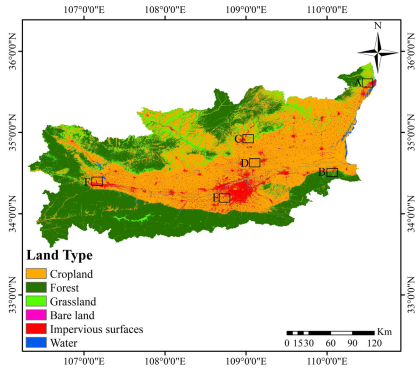


Fig. 6. Land-cover type distribution map of the GZ in 2020.

spread from east to west and north to south. The period from 1990 to 2022 was segmented into eight distinct periods; the number of change pixels within each interval statistically was assessed using ArcGIS. A "rise–stabilize–rise" trajectory was observed in land-cover modifications across the GZ: initial increase from 1990 to 1998, stable between 1998 and 2014, and pronounced growth from 2014 to 2022.

D. Overview and Changes in Land-Cover Type Distributions in the GZ

Fig. 6 showcases the land-cover distribution in the GZ in 2020, dominated by cropland and forest. Impervious surfaces and grassland were also present but less prevalent, and water and bare land were the least common. Cropland primarily spanned the central plains and to a lesser extent the northern plateau, while forests were concentrated in the southern Qinling Mountains and the northeastern parts of Baoji and Xianyang. Grasslands, which are naturally fragmented, were found in the northern plateau, and impervious surfaces fragmented and scattered were distributed in the central plains. Moreover, water and bare land were distributed in linear and dotted patterns in the central plains.

Fig. 7(a) illustrates the spatial distribution of land-cover change intensity within the GZ from 1990 to 2022, with a

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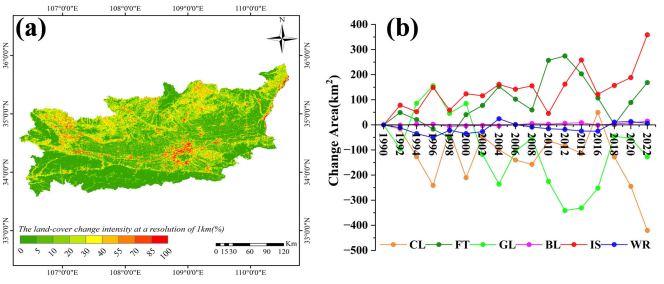


Fig. 7. (a) Spatial distribution of land-cover change intensity in the GZ from 1990 to 2022. (b) Net area changes of six land-cover types in the GZ from 1990 to 2022. CL: cropland, FT: forest, GL: grassland, BL: bare land, IS: impervious surfaces, and WR: water.

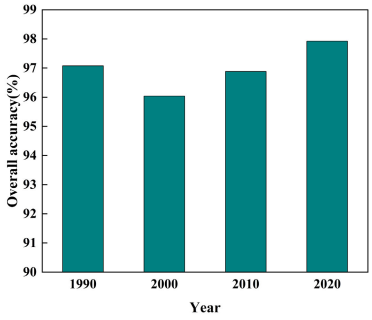


Fig. 8. Overall accuracy of time-series land-cover based on independent validation samples.

1-km spatial resolution grid used to measure the proportion of changed pixels. Over the past 32 years, the area has experienced considerable land-cover transformations, predominantly in three regions: 1) river interflow zones around the Yellow River and Wei River channels, where seasonal river fluctuations have prompted shifts from cropland to water; 2) the northern plateau, particularly the Weibei Plateau, where variations in rainfall and soil composition have facilitated transitions among forest, cropland, and grassland; and 3) urban and residential sectors, notably the urban expanse of Xi'an within the central plains, where urban growth has accelerated, representing a significant driver of change from cropland to impervious surfaces.

Fig. 7(b) presents a quantitative analysis of the net changes in six types of land-cover within the GZ from 1990 to 2022. A notable inverse relationship was observed in the trends among cropland, impervious surfaces, forest, and grassland areas. Specifically, a decreasing trend occurred in the areas dedicated to cropland, grassland, and water bodies and an increasing trend in forest land, impervious surfaces, and bare land. The most substantial reduction occurred in cropland, which decreased by ∼1396.95 km^2, followed by grassland (∼1091.58 km^2) and water bodies (∼146.76 km^2). Conversely, the area of impervious surfaces experienced the most significant growth and expanded by ∼1574.11 km^2, followed by forest areas (∼1037.41 km^2) and bare land with a slight increase (∼23.77 km^2).

Fig. S10 presents the net changes in six land-cover types within five cities of the GZ from 1990 to 2022, revealing diverse transformation patterns. Notably, shifts in cropland, forested areas, grasslands, and impervious surfaces were prominent. In Xi'an, Xianyang, and Baoji, cropland consistently decreased, with Xi'an experiencing the most pronounced decrease of ∼621.87 km^2. Tongchuan City exhibited a period of dynamic balance from 1990 to 2002 and subsequently entered a phase of gradual expansion, experiencing an increase of ∼131.63 km^2 over the three decades. The cropland pattern in Weinan City oscillated, displaying a trend of reduction, expansion, and reduction, with an overall stable area. The forest areas in Xi'an, Xianyang, and Baoji demonstrated a marked "J"-shaped growth pattern, with a significantly larger trend post-2000. In contrast, Tongchuan and Weinan exhibited an "n"-shaped fluctuation in forest cover. Grasslands, particularly in Baoji, significantly decreased by ∼432.48 km^2. Conversely, impervious surfaces, especially in Xi'an, increased by ∼570.54 km^2.

The Sankey diagram in Fig. S11 effectively visualizes the land-cover transition dynamics within the GZ across four distinct periods from 1990 to 2022. Substantial shifts occurred in cropland, forest, grassland, and impervious surfaces compositions. Notably, cropland and grassland experienced a considerable decline, with their respective percentages decreasing from 47.04% and 10.4% in 1990 to 44.53% and 8.43% in 2022, respectively. The reduction in cropland was primarily absorbed by the expansion of impervious surfaces, grassland, and forest, with a notable increase in conversion to impervious surfaces over time. Conversely, the reduction in grassland was largely reallocated to cropland and forest areas. Forest and impervious surface areas witnessed a net gain from 38.02% and 3.56% in 1990 to 39.89% and 6.39% in 2022, respectively. Interestingly, forest areas predominantly transitioned into grassland, whereas cropland primarily transitioned into impervious surfaces.

Fig. S12 provides a visual representation of the land-cover transformations within Xi'an city in the GZ over the last 32 years, highlighting the impact of urbanization. This period has seen a significant conversion of cropland into impervious surfaces, amplifying the rate and intensity of land-cover changes within the city. Notably, urban expansion was markedly more rapid before 2014, characterized by a northward extension. This dataset, synthesized through the CCD_EAVI method and leveraging dense temporal remote sensing data, adeptly captures the spatiotemporal dynamics of surface changes providing a foundation for advancing research into land use evolution within the GZ and underscoring the significant impact of urbanization on the region's land cover.

E. Accuracy Assessment of the Land-Cover Map of the GZ

Table S4 showcases the confusion matrix for land-cover classification results derived from a visual interpretation of independent validation samples in 2020. The land-cover classification in the GZ for that year achieved an overall accuracy of 97.92%. This was consistent across land-cover types, including cropland, forest, water, grassland, and impervious surfaces, for which the PA, UA, and F1-score exceeded 92%. Despite the scarcity and fragmentation of bare land within the region, the PA, UA, and F1-score surpassed 82%.

Fig. 8 presents a per-pixel accuracy evaluation of the classification results utilizing an independent annual validation sample set. The classification accuracy remained notably high across different years, consistently exceeding 96%. Notably,

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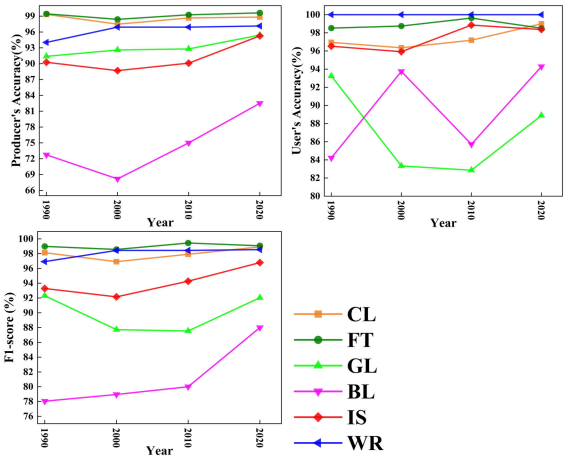


Fig. 9. Time-series mapping accuracy of PA and UA for six land-cover types based on independent validation samples. CL: cropland, FT: forest, GL: grassland, BL: bare land, IS: impervious surfaces, and WR: water.

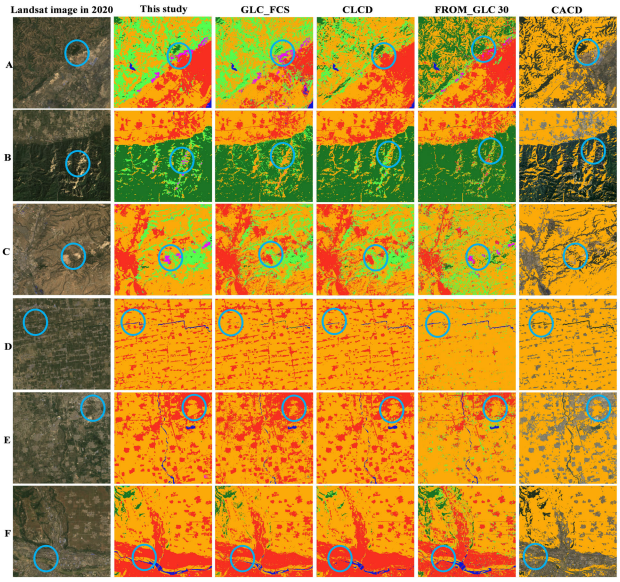


Fig. 10. Detailed comparison between this study and multisource land-cover products. Columns show the details of this study's 2020 product compared to the 30-m resolution 2020 products GLC_FCS, CLCD, CACD, and 30-m resolution 2017 product FROM_GLC30. Rows show six different sites from Fig. 6; land-cover type color legends are the same as in Fig. 6. Detail differences between this study and multisource land-cover products are marked with blue circles.

2020 marked the peak in mapping accuracy, potentially attributable to the enhanced quantity and quality of Landsat 8/9 imagery (Fig. S1). The accuracy metrics for cropland, forest, and water maintained stable and high accuracy rates >94% across various years (Fig. 9). Due to the complexity of impervious surface landscapes and the high fragmentation of grasslands, primarily located in arid plateau areas susceptible to climate change, the classification accuracy was relatively reduced. Nevertheless, PA, UA, and F1-score remained >82%. The lowest F1-score for bare land in 1990 was 78.05%.

F. Comparison With Other Products

Fig. 10 compares the spatial distribution of land-cover in six selected areas within the GZ using the CCD_EAVI product against three land-cover products of differing resolutions (Fig. 6). Given the extensive cropland-cover age in the region, a comparison with the CACD product, specifically for cropland detail, was also performed. Relative to the 30-m resolution land-cover products (GLC_FCS and CLCD), the method proposed in this study demonstrates high consistency in spatial distribution and considerable improvements in spatial detail and classification accuracy, exhibiting less noise and smaller patch occurrence. For instance, in areas Fig. 10(A) and Fig. 10(C), both comparative products incorrectly identified bare land as impervious surfaces; in area Fig. 10(B), they erroneously classified mountaintop bare land and grassland as cropland; in area Fig. 10(D), they overlooked certain small water areas; and in area Fig. 10(E), cropland adjacent to rivers was erroneously classified as impervious surfaces. Relative to the 30-m spatial resolution FROM_GLC30 product, the CCD_EAVI product rectified several misclassifications, such as forest and grassland in area Fig. 10(A), bare land and impervious surfaces in area Fig. 10(B), and impervious surfaces and grassland in areas Fig. 10(D) and Fig. 10(E). Relative to the CACD product, the proposed product more accurately estimated the cropland area within the GZ. For example, in region Fig. 10(B), the CACD product misclassified bare land and grassland as cropland, while in region Fig. 10(C), it classified bare land as cropland.

V. Discussion

A. Advantages of the CCD_EAVI in Tracking Land-Cover Changes

The CCD algorithm leverages continuous satellite imagery observations for fixed-order harmonic fitting, generating harmonic function predictions for specific time points to compare with actual observations [24]. This method boasts automation, broad applicability, and minimal restrictions on data sources. However, its reliance on a singular threshold often results in substantial omission errors when detecting land-cover changes over extensive areas. This limitation becomes particularly pronounced in the face of short-term, minor, and complex land-cover alterations, where a single linear model cannot accurately simulate change trajectories. This inadequacy is attributed to the seasonality and nonlinearity inherent in vegetation dynamics, which complicates the detection of subtle land-cover changes. At the pixel scale, which is influenced by the imagery resolution, complete land-cover transformations may not be captured within a single pixel, necessitating further image processing and refinement for accuracy. For instance, the CCDC_SMA model represents the integration of SMA with the CCDC algorithm, and it has effectively tracked forest degradation and logging activities overextended periods in Georgia, achieving a mapping accuracy of up to 91% [29]. Moreover, enhancing and magnifying the differences between pixels can further reduce the impact of background noise in the image. For instance, the CCDC_PFI model developed by Cai et al. [30] focuses on magnifying the disparities between forest and nonforest pixels beyond the capabilities of the CCDC_SMA model; however, it overlooks the absolute error in vegetation indices between standard and detected endmembers. Land-cover change detection presents

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TABLE II
ACCURACY ASSESSMENTS OF THE ESTIMATED YEAR OF THE FIRST LAND-COVER CHANGE

	Deviation between estimation and reference		
Year(s)	0	1	>1
Proportion (%)	94.00	3.33	2.67
Mean deviation	0.07 years		
Mean absolution deviation	0.13 years		

complexities surpassing those encountered in forest change detection. The newly introduced EAVI transcends traditional change detection metrics by accounting for the stochasticity and seasonal aspects of land cover, significantly mitigating random errors (Figs. 3 and S5). Moreover, it incorporates endmember abundance, optimal spectral bands for land surface sensitivity, and spectral index features and can, thus, effectively distinguish between various land-cover types (Fig. 6). In conjunction with the CCD algorithm, the EAVI enables the swift and automated documentation of a wide array of land-cover and land use changes, encompassing disturbances and modifications (Table II and Fig. S4) and offering a more comprehensive and nuanced approach to tracking environmental changes.

B. Strengths and Potential Implications

The advent of big data platforms and the availability of medium-resolution remote sensing imagery have revolutionized large-area land-cover mapping. However, collecting and updating representative training data remain challenging due to costs and complexities. This study introduces a novel method integrating SMA with adaptive strategies for collecting high-quality training samples to automate land-cover mapping in five key steps: 1) utilizing the SMA method to create EAVI indicators, enhancing land-cover spectral feature distinctions; 2) merging multisource land-cover products to identify time-stable regions and employing the HDI_{RMSE} indicator to detect internal region heterogeneity, thereby refining sample quality; 3) automatically updating the training sample dataset using a hexagonal adaptive strategy, thereby improving its representativeness; 4) leveraging the CCD model and a comprehensive Landsat image series to identify change pixels at any moment, facilitating the automated refresh of samples; and 5) ensuring the stability and accuracy of time-series classification products through temporal and spatial consistency verification.

An optimized feature space and a well-curated training sample set significantly boost machine learning algorithm efficiency, reduce computational demands, and minimize overfitting risks [31]. The EAVI feature set, sensitive to land-cover surfaces, requires no further feature selection, thus streamlining computations. The training pool, derived from multisource, multitemporal land-cover products and purified with the HDI_{RMSE} indicator, minimizes mixed pixels at stable region boundaries (Fig. 5), enhancing sample stability and reducing the historical misclassifications of single product [38] (Fig. 10). Increasing the number of samples can improve model accuracy, but ineffective samples may reduce this benefit. After accumulating 7500 sample points, a decrease in classification accuracy was observed (Fig. S8). To address this, 1500 random samples were added using a hexagonal adaptive sampling strategy, primarily in urban areas, the arid northern highlands, and other regions with high land change intensity and fragmentation (Figs. 7 and S13). This expanded the intersection of six land-class sample clusters, enhancing overall classification accuracy by 1.2% (Table S4 and Fig. S8). Hence, adaptive sampling effectively enhances the representation of samples in fragmented regions. Furthermore, combining spatiotemporal consistency optimization algorithms can further improve the reliability and spatial continuity of land-cover classification (Fig. S14). Compared to traditional methods relying on spectral distance and similarity for training sample updates [16], the highly automated CCD algorithm captures pixel changes at any time with over 96% mapping accuracy for updated training samples (Fig. 8), advancing land-cover mapping automation.

C. Drivers of Land-Cover Changes Within the GZ

Fig. 7 highlights the significant land-cover changes within the GZ, showcasing variations in intensity influenced by climate conditions, economic development, and policy decisions. The relationship between these factors and land-cover change intensity is multifaceted.

1) *Climate Influence:* Land-cover change intensity shows an inverse correlation with rainfall, especially in the plateau regions of the northern GZ. Here, the predominant land covers are cropland and grassland, which are shaped by topographical and soil conditions, and agriculture is reliant on natural rainfall. With the recent high-temperature droughts and decreased rainfall, cropland abandonment has increased. Furthermore, the Yellow River and Wei River, characterized as seasonal rivers, demonstrate significant annual spatial variations in water distribution attributable to seasonal rainfall disparities.
2) *Economic Development:* The intensity of land-cover change directly correlates with economic growth rates. This is particularly evident in Xi'an, the provincial capital, where the extent of land-cover change is notably high. Fig. S12 further corroborates that urbanization is a primary driver of the conversion from cropland to impervious surfaces, highlighting the impact of economic advancement on land-use dynamics.
3) *Policy Impact:* The significant increase in forested areas in the GZ post-2000 can be attributed to the "Returning Farmland to Forest" project initiated by the Chinese government in Yan'an, Shaanxi, in 1999, followed by the widespread application of related policies throughout the province.

D. Limitations and Outlook of the Method

While the proposed method efficiency automates training sample generation for time-series land-cover mapping, it has certain limitations. First, the HDI_{RMSE} indicator's effectiveness is constrained by the spatial resolution of Landsat imagery, potentially overlooking minor features like field roads smaller

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than a pixel, resulting in mixed pixels within the training sample pool. Second, the CCD algorithm's efficacy depends on the availability of dense, cloud-free satellite observations, with reliability tied to the density of effective imagery [24], [25]. Given the reliance on Landsat data, especially pre-1999 when only Landsat 5 was available, image sparsity in earlier years could hinder accurate land-cover change detection (Fig. S1). Third, the study's scope was restricted to evaluate the RF classifier's performance in pixel-scale land-cover mapping. Future research could explore advanced algorithms based on deep learning, such as convolutional neural networks (CNNs) and long short-term memory (LSTM) networks and object-oriented approaches, which might enhance mapping accuracy. Future studies could also utilize higher resolution optical imagery, including Sentinel-2 and GF satellites, to gather a denser series of remote sensing images, improving spatial and temporal granularity. Moreover, integrating object-oriented methods with deep learning techniques can potentially refine land-cover mapping accuracy and efficiency. The current study's assessment of the GZ suggests that the proposed approach could scale to larger, possibly global extents for long-term land-cover monitoring, advancing remote sensing and land-cover analysis.

VI. Conclusion

This research developed a novel framework for land-cover mapping on the GEE platform, aimed at enhancing the timeliness and accuracy of long-term time-series land-cover dynamic monitoring, crucial for understanding global climate change drivers. The primary study outcomes include the 1) development of a new EAVI through SMA that outperforms traditional methods in identifying small patches and assessing change integrity; 2) creation of the HDI_{RMSE} indicator to detect internal regional heterogeneity, enabling the automatic generation of a training sample pool from multisource land-cover products; 3) use of a hexagonal adaptive strategy to automatically enhance the representativeness of the training sample dataset; 4) application of the CCD_EAVI model for automated sample updates across years by identifying stable regions in the time series; and 5) use of a temporal and spatial consistency algorithm to optimize time-series classification products. The accuracy evaluation demonstrated exceptional results, with the overall accuracy of land-cover mapping of the GZ in 2020 reaching 97.92%, and an average mapping accuracy exceeding 96% over the past 32 years across independent validation samples. This method exhibited superior spatial detail, reduced noise, and enhanced classification accuracy compared to benchmark products. Hence, this study presents a robust approach for the automated generation of large-scale samples and time-series land-cover mapping in arid regions, significantly contributing to environmental monitoring and climate change research.

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